8. GOODWILL

Goodwill allocated to reportable segments as of December 31, 2019 and 2018 and changes in the carrying amount of goodwill during the years ended December 31, 2019 and 2018 are as follows:

	Automotive and Discrete Group (ADG)	Microcontrollers and Digital ICs Group (MDG)	Analog, MEMS & Sensors Group (AMS)	Total
December 31, 2017	**—**	**121**	**2**	**123**
Foreign currency translation	—	(2)	—	(2)
December 31, 2018	**—**	**119**	**2**	**121**
Business combination	43	—	—	43
Foreign currency translation	—	(2)	—	(2)
December 31, 2019	**43**	**117**	**2**	**162**

As described in Note 7, the acquisition of Norstel resulted in the recognition of $43 million in goodwill which has been included in the ADG segment to align the goodwill of the acquired Company with the segment under which the related activities will be reported.

As of the end of the third quarters of 2019 and 2018, the Company performed its annual impairment test. The Company did not elect to perform a qualitative assessment. The impairment test was conducted following a two-step process. In the first step, the Company compared the fair value of the reporting units tested to their carrying value. Based upon the first step of the goodwill impairment test, no impairment was recorded since the fair value of the reporting units exceeded their carrying value.

Goodwill as at December 31, 2019 and 2018 is net of accumulated impairment losses of $102 million, of which $96 million relates to the MDG segment and $6 million to Others. In 2019, 2018 and 2017, no impairment loss was recorded by the Company.

9. OTHER INTANGIBLE ASSETS

Other intangible assets consisted of the following:

December 31, 2019	Gross Cost	Accumulated Amortization	Net Cost
Technologies & licenses	699	(578)	121
Purchased & internally developed software	486	(427)	59
Technologies in progress	119	—	119
Other intangible assets	70	(70)	—
Total	**1,374**	**(1,075)**	**299**

December 31, 2018	Gross Cost	Accumulated Amortization	Net Cost
Technologies & licenses	705	(592)	113
Purchased & internally developed software	459	(404)	55
Technologies in progress	44	—	44
Other intangible assets	69	(69)	—
Total	**1,277**	**(1,065)**	**212**

As described in Note 7, the acquisition of Norstel resulted in the recognition of technology in process for $86 million in the line “Technologies in progress”.

The line “Technologies in progress” in the table above also includes internally developed software under construction and software not ready for use.

The amortization expense in 2019, 2018 and 2017 was $69 million, $64 million and $58 million, respectively.